TU CSIT: Data Structures & Algorithms (DSA) Strategic Study Guide

This guide is based on the analysis of question papers from 2074 to 2081.

🚀 The "Big Three" Units (42% of Total Marks)

If you are short on time, focus on these three units first. They consistently provide the long questions (10 Marks).

1. **Unit 8: Trees and Graphs (15.0 Avg Marks)**
   - **Focus:** AVL Rotations (LL, RR, LR, RL), BST Insertion/Deletion, and MST Algorithms (Kruskal/Prim).
   - **Tip:** Always practice "Tracing" graphs for Dijkstra's and MST; the theory is rarely asked without a numerical/graph trace.
2. **Unit 6: Sorting (12.5 Avg Marks)**
   - **Focus:** Quick Sort (Divide & Conquer) is the most frequent 10-pointer. Heap Sort (Building Max Heap) is the second favorite.
   - **Tip:** Learn the partition algorithm for Quick Sort by heart.
3. **Unit 5: Lists (10.0 Avg Marks)**
   - **Focus:** Doubly Linked List (Insert/Delete) and Singly Linked List operations.
   - **Application:** Implementation of Stack/Queue using Linked Lists.

🎯 The "Sure-Shot" 10-Mark Questions

Historically, the 3 long questions usually come from this pool:

1. **Infix to Postfix Conversion:** Algorithm + Manual Trace (Unit 2).
2. **Quick Sort or Merge Sort:** Algorithm + Step-by-step Trace (Unit 6).
3. **Binary Search Tree (BST) or AVL Tree:** Insertion/Deletion logic or Rotation examples (Unit 8).
4. **Linked List Operations:** Complete algorithms for Doubly or Singly lists (Unit 5).

📝 The "High-Frequency" 5-Mark Questions

These topics appear almost every alternate year:

- **Unit 1:** Asymptotic Notations (O, Ω, Θ) and malloc vs calloc.
- **Unit 3:** Circular Queue algorithms (How they solve Linear Queue limitations).
- **Unit 4:** Tower of Hanoi (TOH) trace for 3 disks.
- **Unit 7:** Collision Resolution techniques (Linear Probing vs. Chaining).

💡 Pro-Tips for the Exam

- **Algorithm Writing:** Don't just write C code. Write clear, numbered steps (Pseudocode). TU examiners prefer structured logic.
- **Tracing Tables:** For Stack (Postfix evaluation) and Sorting, always use a table format to show the state of the data structure at each step.

- **Complexity:** For every sorting and searching algorithm you learn, memorize its Best, Average, and Worst-case time complexity.
- **Short Notes (Q12):** These are "free marks." Common topics include: *Abstract Data Type (ADT), Priority Queue, Tail Recursion, and Hashing.*